Bin-Resolution-Independent Radiative Flux Mapping and Assessment for Cylindrical Solar Receivers

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Supplementary Information

S1. Trapezoidal integration of the ASTM G173-03 spectrum

The cumulative spectral irradiance function $G_{\text{cum}}(\lambda)$ is evaluated numerically from the discretised ASTM G173-03 tabulated spectrum $\{(\lambda_i, G_{b,i})\}$ using the trapezoidal rule,

$$G_{\text{cum}}(\lambda_i) = G_{\text{cum}}(\lambda_{i-1}) + \tfrac{1}{2}(G_{b,i} + G_{b,i-1})(\lambda_i - \lambda_{i-1}), \qquad G_{\text{cum}}(\lambda_0) = 0 \tag{1}$$

This initializes the cumulative sum at zero and accumulates trapezoidal areas between consecutive spectral points. The band clipping edges a_j and b_j for each wavelength band λ_j are

$$a_j = \max\!\left(\lambda_j - \tfrac{\Delta\lambda}{2},\ \lambda_{\min}\right), \qquad b_j = \min\!\left(\lambda_j + \tfrac{\Delta\lambda}{2},\ \lambda_{\max}\right) \tag{2}$$

The in-band irradiance is given by $\Delta G_j = G_{\text{cum}}(b_j) - G_{\text{cum}}(a_j)$.

S2. Theoretical noise scaling of the three reconstruction methods

The three methods exhibit fundamentally different noise scaling with ray count and grid resolution.

For the bin-based method, the expected number of rays per bin of area ΔA_{grid} is $\bar{n} = N_{\text{rays}}/(N_\theta N_s)$. The relative noise scales as

$$\text{Noise}\%_{\text{BB}} \propto \frac{1}{\sqrt{\bar{n}}} = \sqrt{\frac{N_\theta N_s}{N_{\text{rays}}}} \tag{3}$$

Grid refinement at fixed N_{rays} amplifies noise — the resolution-noise coupling at the root of the checkerboard problem.

For the k-NN estimator, exactly k rays contribute to every evaluation point by construction,

$$\text{Noise}\%_{\text{kNN}} \propto \frac{1}{\sqrt{k}} \tag{4}$$

This is independent of evaluation grid resolution, motivating the k-optimization study of Section 5.2 of the main manuscript.

For the KDE estimator with Scott's bandwidth $h \propto N_{\text{rays}}^{-1/6}$, the effective sample size within the kernel footprint scales as $N_{\text{rays}}h^2$, giving

$$\text{Noise}\%_{\text{KDE}} \propto \frac{1}{\sqrt{N_{\text{rays}}\, h^2}} \propto N_{\text{rays}}^{-1/3} \tag{5}$$

Here $h = h_{\text{eff}} = h_{\text{Scott}} \cdot \sqrt{\hat{\sigma}_c \hat{\sigma}_s}$ is the geometric-mean physical bandwidth computed after inverse normalization.

This is both grid-independent and self-adjusting: the noise floor contracts automatically as more rays are simulated, without any user intervention.

S3. FFTKDE cross-verification against `scipy.stats.gaussian_kde`

Table 1 reports the percentage difference in peak flux between FFTKDE and the direct-summation scipy reference implementation at matched N_λ and grid resolution. Positive values indicate FFTKDE exceeds scipy. Mean flux agreement between the two implementations is within 0.01% in all cases.

Table 1: Percentage difference in peak flux between FFTKDE and the direct-summation `scipy.stats.gaussian_kde` reference, at $N_\lambda \in [500, 10000]$ rays per wavelength band. All KDE results for $N_\lambda > 10000$ use FFTKDE exclusively.

Grid	$N_\lambda = 500$	1000	2000	5000	10000
20×20	0.87	0.02	0.68	1.50	0.63
80×80	1.98	2.77	1.32	1.45	1.21
300×300	1.96	2.47	1.93	1.71	1.23
60×400	2.20	2.58	1.64	1.78	1.47
80×600	1.99	2.80	1.98	1.81	1.54
100×1000	1.95	2.57	1.93	2.15	1.64

Mean absolute difference at $N_\lambda = 10000$: 1.3% (range 0.63–1.64%).
Maximum across all tested N_λ: 2.80%.

The deviation arises because FFTKDE evaluates the continuous Gaussian density on an internal 1024^2 grid and interpolates to the export centroids, whereas scipy evaluates the kernel sum exactly at each centroid. This is a bounded, monotonically decreasing discretization effect (confirmed by the internal resolution study in S4) that is smaller than the run-to-run Monte Carlo variability of the peak cell (2–4 kW/m^2, Section 5.2 of the main manuscript and Supplementary S9) and does not affect any conclusion of the study.

S4. FFTKDE internal FFT bin resolution study

Table 2 reports the sensitivity of the reconstructed peak and mean flux to the internal FFT resolution size Γ at the 60×400 export resolution and $N_\lambda = 10000$. The study confirms that $\Gamma = 1024^2$ provides adequate discretization of the continuous density field for the present application.

The peak flux varies monotonically and by less than 0.9% across the full range $\Gamma \in \{512^2, 2048^2\}$. The residual discrepancy between $\Gamma = 2048^2$ and the scipy direct-summation reference (1.0%) arises because FFTKDE interpolates from the internal grid to the export centroids, whereas scipy evaluates the kernel sum exactly at each centroid. This is a bounded, monotonically decreasing discretization effect. The peak flux discrepancy between $\Gamma = 1024^2$ and the scipy reference is $(45069 - 44416) = 653\,\text{W/m}^2$, equivalent to

Table 2: FFTKDE internal resolution study at the 60×400 export mesh, $N_\lambda = 10000$, $N_{\text{rays}} = 1118896$. Mean flux is conserved to machine precision at all grid sizes after energy-closure rescaling.

Γ	$q_{\max}$ (W/m^2)	q_{mean} (W/m^2)	Time (s)
512^2	45262	18765	3.3
1024^2	45069	18765	3.4
2048^2	44874	18765	4.0
scipy (ref.)	44416	18765	1196

1.47%. This is smaller than the natural run-to-run Monte Carlo variability of the peak cell (2000–4000 W/m^2, demonstrated in Section 5.2 of the main manuscript through the ray-count convergence study), confirming that the internal grid discretization is not the dominant source of uncertainty in the reported peak flux values. At the working ray count $N_\lambda = 10000$, the FFTKDE peak flux at $\Gamma = 1024^2$ exceeds the scipy reference by 0.6–1.6% depending on export mesh resolution (Table 1 in Supplementary S3), confirming that this is a bounded discretization effect rather than a systematic error.

S5. FTP composite score sensitivity to component weighting

Figure 1 shows the FTP composite score for all matched-resolution cases under three weight schemes: V1 (equal, $v_i = 0.20$), V2 (SSI-dominant, $v_{\text{SSI}} = 0.50$), and V3 (accuracy-dominant, $v_{\text{ETA}} + v_{\text{PTA}} = 0.60$). The method ranking BB $<$ k-NN $\approx$ KDE is preserved under all three schemes. BB deficits are distributed simultaneously across PTA, SSI, and RCR; no single-metric reweighting eliminates all three.

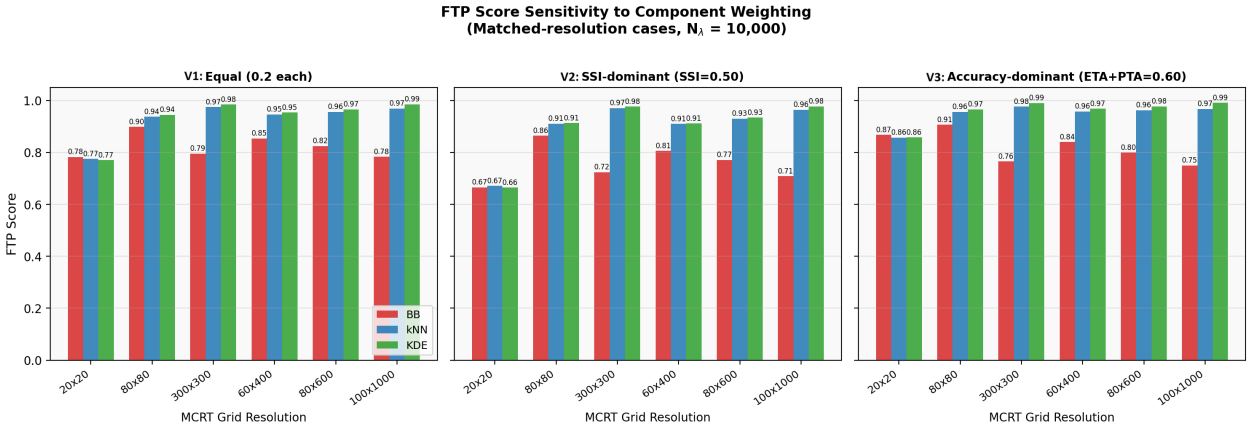


Figure 1: FTP composite score for matched-resolution cases at $N_\lambda = 10000$ under three weight schemes. Red: BB; blue: k-NN; green: KDE. The BB/(k-NN, KDE) ranking is stable across all schemes.

Table 3 provides the numerical values of the weight schemes and resulting mean FTP scores for completeness.

Table 3: Alternative FTP weight schemes and resulting method mean FTP scores across the six matched-resolution cases at $N_\lambda = 10000$.

Scheme	v_{ETA}	v_{PTA}	v_{SSI}	v_{ILR}	v_{RCR}	BB	k-NN	KDE
V1 (equal)	0.20	0.20	0.20	0.20	0.20	0.770	0.930	0.934
V2 (SSI= 0.50)	0.10	0.10	0.50	0.10	0.20	0.681	0.921	0.928
V3 (ETA+PTA= 0.60)	0.30	0.30	0.15	0.15	0.10	0.735	0.927	0.944

S6. Sensitivity of FTP scores to $q_{\mathrm{max,ref}}$

Table 4 reports the mean composite FTP score and the mean Peak Transfer Accuracy (PTA) across the six matched-resolution cases for three physically motivated values of the reference peak flux $q_{\mathrm{max,ref}}$. PTA is defined as $q_{\mathrm{max}}^{\mathrm{final}}/q_{\mathrm{max,ref}}$ and is therefore proportional to the reference value; the composite FTP inherits this sensitivity through PTA. Other parameters like ETA, SSI, ILR, and RCR do not involve $q_{\mathrm{max,ref}}$ and are invariant to this parameter.

Table 4: Sensitivity of mean matched-resolution FTP score and mean PTA to $q_{\mathrm{max,ref}}$. Values are means across all six matched-resolution cases at $N_\lambda = 10000$.

	Mean FTP score			Mean PTA (%)		
$q_{\mathrm{max,ref}}$ (W/m^2)	BB	k-NN	KDE	BB	k-NN	KDE
40000 [a]	0.755	0.903	0.914	193.8	115.7	110.6
44500 [b]	0.772	0.926	0.934	174.2	104.0	99.5
49000 [c]	0.784	0.924	0.916	158.2	94.5	90.3

[a] [Goyal and Reddy(2022)] $\mathrm{LCR}_{\mathrm{max}} \approx 46$ at DNI = 900 W /m^2.
[b] [Zhu et al.(2023)] $\mathrm{LCR}_{\mathrm{max}} \approx 51$ at DNI = 860 W/m^2.
[c] [Qiu et al.(2017)] $\mathrm{LCR}_{\mathrm{max}} \approx 54$ at DNI = 900 W/m^2.

The method ranking BB $<$ k-NN $\approx$ KDE is preserved across all three reference values. The maximum shift in composite FTP score across the full range is 0.018 points. At the adopted central value of 44500 W/m^2, the mean PTA for k-NN (104.0%) and KDE (99.5%) confirms that both density-estimation methods recover the physical peak with less than 5% deviation. BB returns 174.2% at the same reference owing to Monte Carlo noise spikes in underdense fine bins – a property of the reconstruction method rather than the reference choice. The range $q_{\mathrm{max,ref}} \in [40000, 49000]$ W/m^2 is taken for study as the physically plausible span for the LS-3 PTC under the optical and spectral conditions of this study. Values below 40000 W/m^2 would imply a peak-to-average ratio below 2.1, inconsistent with the concentration geometry of the LS-3 collector ($C_{\mathrm{geom}} = 26.2$, rim angle 79.2°). Values above 49000 W/m^2 correspond to idealized specular models with no circumsolar scattering, which overestimate the peak for realistic Buie-model conditions. The sensitivity analysis is therefore complete over the physically relevant domain, and the conclusions are not conditional on the specific value adopted within this range.

S7. FTP composite score with and without the Resolution Compatibility Ratio

The Resolution Compatibility Ratio (RCR) is a purely geometric metric that quantifies the match between the MCRT source bin-resolution and the target CFD mesh resolution, together with an absolute penalty for face sizes comparable to the receiver diameter. By construction, RCR is independent of the flux field values and evaluates only the mesh topology. Its inclusion in the composite FTP score with equal weight $v_i = 0.20$ has been questioned because it may structurally favor density-estimation methods, which can evaluate directly at any mesh centroid, over the bin-based method, which is constrained to its source bin resolution. The 4-metric composite reported in this section removes RCR entirely and redistributes its weight equally among the four field-quality metrics (ETA, PTA, SSI, ILR), each receiving $v_i = 0.25$. This provides an assessment of whether the observed method ranking depends on the inclusion of a geometric metric or emerges purely from the flux field quality characteristics of each method.

Table 5 compares the 5-metric composite FTP score (including RCR, equal weight $v_i = 0.20$) with the 4-metric composite obtained by removing RCR and redistributing weight equally among the remaining four metrics ($v_i = 0.25$). Results are for the matched-resolution cases at $N_\lambda = 10000$.

Table 5: Mean matched-resolution FTP composite score with and without the Resolution Compatibility Ratio (RCR). Removing RCR widens rather than narrows the gap between BB and the density-estimation methods.

Composite	BB	k-NN	KDE
5-metric (with RCR, $v_i = 0.20$)	0.772	0.926	0.934
4-metric (without RCR, $v_i = 0.25$)	0.759	0.952	0.962

Removing RCR increases the k-NN and KDE scores (because the four remaining metrics all favor density-estimation methods) and decreases the BB score (because RCR was providing partial credit to BB matched-resolution cases). This result confirms that RCR does not artificially inflate the advantage of density-estimation methods; rather, it slightly attenuates it by rewarding BB for the coincidental resolution match in the diagonal cases.

S8. Sensitivity of k-NN FTP scores to neighborhood size k

Table 6 reports FTP component metrics and the composite score at the matched-resolution 60×400 case for $k \in \{300, 500, 700\}$ at $N_\lambda = 10000$.

The composite FTP score varies by 0.019 across the tested range, representing approximately one-seventh of the BB-to-k-NN gap of 0.134 at the matched 60×400 case. As k increases, bandwidth smoothing increases, suppressing the reconstructed peak flux toward the physical reference and reducing SSI. The value $k = 500$ is selected as the inflection point of the $q_{\max}$ versus k convergence curve (Section 5.2), balancing noise suppression against peak resolution. Values above $k = 700$ produce visibly over-smoothed spatial flux maps in which genuine circumferential gradients are attenuated.

Table 6: k-NN FTP metrics at the matched-resolution 60×400 case ($N_\lambda = 10000$) for three neighborhood sizes. The k-NN composite score exceeds BB (FTP = 0.812 at this resolution) in all cases.

k	FTP composite	PTA (%)	SSI (%)	$q_{\max,\mathrm{kNN}}$ (W/m^2)
300	0.935	109.1	14.5	52992
500	0.946	104.4	13.7	47544
700	0.954	100.5	13.2	46119

S9. Run-to-run Monte Carlo uncertainty

To assess the run-to-run variability of the reported FTP scores, the complete reconstruction and assessment pipeline was repeated for three additional independent MCRT realizations at $N_\lambda = 10000$ using different system-clock random seeds (morning, afternoon, and evening runs on identical hardware). All realizations used the same geometry, material properties, sun position, and spectral irradiance model. The total absorbed power P_{MCRT} varied by less than 0.2% across the four realizations.

Table 7 reports the individual FTP composite score, PTA, and SSI for each realization and method at the matched-resolution 60×400 case, together with the mean and standard deviation across the four runs.

Table 7: Run-to-run variability of FTP metrics at the matched-resolution 60×400 case across four independent MCRT realizations at $N_\lambda = 10000$. Run 0 is the primary realization used throughout the main manuscript; Runs A–C are independent realizations with different system-clock random seeds.

Method	Run	FTP composite	PTA (%)	SSI (%)
BB	Run 0	0.812	157.2	28.1
	Run A	0.814	156.6	27.8
	Run B	0.832	147.6	27.9
	Run C	0.808	159.5	27.9
	Mean ± std	**0.817 ± 0.010**	**155.2 ± 5.3**	**27.9 ± 0.1**
k-NN	Run 0	0.946	104.4	13.7
	Run A	0.946	104.5	13.7
	Run B	0.947	104.1	13.7
	Run C	0.942	106.4	13.9
	Mean ± std	**0.945 ± 0.002**	**104.9 ± 1.0**	**13.7 ± 0.1**
KDE	Run 0	0.953	99.5	14.2
	Run A	0.953	99.0	14.1
	Run B	0.954	99.8	14.1
	Run C	0.953	99.2	14.4
	Mean ± std	**0.953 ± 0.001**	**99.4 ± 0.3**	**14.2 ± 0.1**

The BB-to-KDE FTP gap of 0.136 points exceeds the BB run-to-run standard deviation by a factor of 13.6, and the BB-to-k-NN gap of 0.128 points by a factor of 12.8. The method ranking BB $<$ k-NN $\approx$ KDE is preserved across all four realizations without exception. The density-estimation methods (k-NN, KDE) show substantially smaller run-to-run variability (standard deviation 0.001–0.002) than the bin-based method (0.010), confirming that continuous density estimators are inherently more stable with respect to Monte Carlo sampling variability than the bin-based approach.

References

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